DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW



Evaluation Scheme & Syllabus

For

B.Tech. 2nd Year

FOOD TECHNOLOGY

(Effective from the Session: 2023-24)

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW

SEMESTER –III

SN	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total SW+ESE	Credit Cr
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BOE3** / BAS303	Science Based Open Elective/BSC (Maths-III/Math IV/ Math V)	T	ES/BS	3	1	0	20	10	30	70	100	4
2	BVE301 / BAS301	Universal Human Value and Professional Ethics/ Technical Communication	T	VA/HS	2	1	0	20	10	30	70	100	3
3	BCH301	Material and Energy Balance	T	PC	3	1	0	20	10	30	70	100	4
4	BFT302	Basics of Food Science and Technology	T	PC	3	1	0	20	10	30	70	100	4
5	BFT303	Basic Food Microbiology	T	PC	2	1	0	20	10	30	70	100	3
6	BFT351	Energy & Environment Lab	P	PC	0	0	2		50	50	50	100	1
7	BFT352	Analysis of Food Constituents Lab	P	PC	0	0	2		50	50	50	100	1
8	BFT353	Basic Food Microbiology Lab	P	PC	0	0	2		50	50	50	100	1
10	BCC301 / BCC302	Cyber Security/Python programming	T	VA	2	0	0	20	10	30	70	100	2
11	BCC351	Internship Assesment /Mini Project*	P							100		100	2
		Total			**15**	**5**	**6**						**25**

- **Mathematics –III** for CE / ENV and allied branches
- **Mathematics-IV** for Computer/Electronics/Electrical & allied Branches, Mechanical & Allied Branches Textile/Chemical & allied Branches
- **Mathematics-V** for Bio Technology / Agriculture Engineering

SEMESTER –IV

SN	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total SW+ESE	Credit Cr
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BAS403 / BOE4**	BSC(Maths-III/Math IV/ Math V)/Science Based Open Elective	T	BS/ES	3	1	0	20	10	30	70	100	4
2	BAS401 / BVE401	Technical Communication / Universal Human Value and Professional Ethics	T	HS/VA	2	1	0	20	10	30	70	100	3
3	BFT401	Fluid Flow and Solid Handling	T	PC	3	1	0	20	10	30	70	100	4
4	BFT402	Food Chemistry	T	PC	3	1	0	20	10	30	70	100	4
5	BFT403	Food Biochemistry & Biotechnology	T	PC	2	1	0	20	10	30	70	100	3
6	BFT451	Fluid Flow & Solid Handling Lab	P	PC	0	0	2		50	50	50	100	1
7	BFT452	Food Chemistry Lab	P	PC	0	0	2		50	50	50	100	1
8	BFT453	Food Biochemistry & Biotechnology Lab	P	PC	0	0	2		50	50	50	100	1
9	BCC402 / BCC401	Python Programming/Cyber Security	P	VA	2	0	0	20	10	30	70	100	2
10	BVE451 / BVE452	Sports and Yoga - II / NSS-II	P	VA	0	0	3			100		100	0
		Total			**15**	**5**	**9**						**23**
		Minor Degree/ Honors Degree MT-1/HT-1											

*****The Mini Project or internship (4 weeks) will be done during summer break after 4th Semester and will be assessed during V semester.

SYLLABUS
III SEMESTER

SUBJECT CODE: BCH301	**COURSE TITLE: MATERIAL & ENERGY BALANCE**
	SEMESTER:III
	PREREQUISITES:NIL

OBJECTIVE:

- To provide basic knowledge of principles of material and energy balances applied to chemical engineering systems and to make them able to calculate the quantity of material input and output of a process plant.

COURSE OUTCOME:

After successful completion of the course, the students will be able to:

- Understand conversions of units, equations, and various concentration measures
- Carry out combustion calculations, proximate and ultimate analysis.
- Apply steady-state and unsteady-state material and energy balance on a system.
- Analyze all the stoichiometric relations being applied to a system undergoing a chemical process and perform the enthalpy calculation.
- Design equipment with inlet and outlet; including recycle-bypass streams for a chemical process and calculate the quantities of utilities required.

REFERENCEBOOKS

S.NO	NAME OFAUTHORS /BOOKS/PUBLISHERS	YEAR OF PUBLICATION /REPRINT
1.	Hougen D.A., Watson K.M. and Ragatz R.A., "Chemical Process Principles", Part-I, 2Ed., CBS Publishers.	1995
2.	Felder R.M. and Rousseradu R.W., "Elementary Principles of Chemical Processes", 3Ed.., John Wiley.	2005
3.	Narayanan K.V. and Lakshmikutty B., "Stoichiometry and Process Calculations", Prentice Hall of India.	2006
4.	Bhatt B.I. and Vora S.M., "Stoichiometry", 5th Ed., TataMcGraw-Hill	2010
5.	Himmelblau D.M. and RiggsthJ. B.," Principles and Calculations in Chemical Engineering", 8Ed., Prentice Hall of India.	2012

Units	Contents	Lecture Hours
I	**Introduction:** Units and dimensions in chemical engineering**,**unit conversion factors, unit conversionofdimensionalequations, Compositions of Mixture and Solutions, stoichiometric and composition relations, Conceptof limiting and excessreactant, conversion, extent ofreaction, selectivity and yield Average molecular weight, density and composition (by weight and by mole) of gas mixture, Ideal and Real Gas Laws - Gas constant - Calculations of Pressure, Volume and Temperature using Ideal Gas Law, Use of Partial Pressure and Pure Component Volume in Gas Calculations, Applications of Real Gas Relationships	**8**

II	**Material Balance & Applications:** Stoichiometric Principles, Concept of material balance, open and closed systems, steady state and unsteady state, multiple component system, selection of a basis,problem solving strategy, Conservation of mass/atom, material lbalance for Systems without chemical reactions involving single unit and multiple units, Material balance for systems with chemical reactions involving single unit and multiple units, analysis of systems with bypass ,recycle and purge streams	**12**
III	**Humidity & Saturation**: Vaporisation & condensation, partial pressure & vapor pressure, vapor-liquid equilibrium and concepts of humidity &saturation, Dry bulb and wet bulb temperature, Calculation of Absolute Humidity, Molal Humidity, Relative Humidity and Percentage Humidity, Use of Humidity in Condensation and Drying, Humidity Chart, Dew Point. Analysis of processes involving condensation, crystallization and vaporization.	**6**
IV	**Energy Balance & Applications:** Conservation of energy, energy balance for systems with and without chemical reactions, Analysis of Products of Combustion of Solid, Liquid and Gas Fuels, Calculation of Excess Air, Heat Capacity of Solids, Liquids, Gases and Solutions, Mean Heat Capacity, Sensible and Latent Heats, Enthalpy, Hess's Law, Standard Heats of Reaction, Formation, combustion,solutionsandmixing,adiabaticandnon-adiabaticreactions, theoretical & actual flame temperatures,chemical engineering problems involving energybalance.	**10**
V	**Degrees of freedom analysis & Unsteady State Material and Energy Balances:** Degrees of freedom analysis for the multi component systems , combined steady state material and energy balance, Unsteady State Material and Energy balances involving with and without chemical reactions.	**6**

SUBJECT CODE: BFT302	**COURSE TITLE:** BASICS OF FOOD SCIENCE & TECHNOLOGY
	SEMESTER: III
	PRE-REQUISITE : NIL

OBJECTIVE: To provide basic knowledge of basic fundamental aspects of Food Science and Technology, Food Composition and Nutritional value, Processing, Preservation and Packaging along with Quality and Safety of Foods to orovide an overview of the operations involved and status of Food Processing Sector in India.

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the scope and status of Food Processing sector in India.
- Understand the Food Constituents , Nutritional value of foods, its estimation and Balanced Diet.
- Gain insight into processing and Preservation of foods and technologies available.
- Gain insight into the Quality aspects of foods and the relevant parameters and their estimation.
- Gain insight into Safety of foods, Safe and Unsafe foods and know the hazards and risks associated with Unsafe foods.

S.No	Name of Authors / Books / Publishers	Year of Publication / Reprint
1	Norman N.Potter, Joseph H.Hotchkiss, " Food Science", 5th Edition, CBS Publishers & Distributors Pvt. Ltd.,	2007
2	Shubhangini A. Joshi, Nutrition and Dietetics, 4th Edition , Tata Mc GrawHill Publishing Company Ltd.	2017
3	WHO Reference Data for Children and Adolescents (5-19 years). Available at http://www.who.int/growthref/en/	2007
4	B.Srilakshmi, Food Science, 7th Edition, New Age International (P) Ltd., New Delhi	2018
5	Kramer, A, Twigg, B A "Quality Control for the Food Industry: Fundamental & Applications, Medtech	2017

COURSE DETAILS:

Units	S.No.	Contents	Lecture Hours
1.	1	Introduction to Food Science & Technology. Food Processing Sector in India: Growth, Opportunities and Challenges. Various types of Food Processing Industries, Market Scenario and Scope. Government Initiatives to promote Food Processing Industry in India.	4
	2	Food and its functions, Physico-chemical properties of foods. Chemical constituents of foods, Desirable and undesirable food constituents and their significance.	5
2.	1	Concept of Food Nutrition, Relation of Nutrition to good health. Food Energy, Determination of energy requirements. Malnutrition: Causes and Effects. Factors affecting the bioavailability of nutrients. Common Nutritional deficiency diseases.	5
	2	Recommended Dietary Allowances (RDA) or Intakes (RDI). Balanced diet and basic food groups . Basal metabolic rate and dietary requirements of different age groups.	4
3.	1	Basic concept of Food Processing and Food Preservation. Familiarization with commonly used terms as Cooking, Roasting, Broiling, Stewing, Simmering, Pasteurization etc. General Principles of Food Preservation. Name of Methods Employed for Food Preservation.	4
	2.	Basic idea of Functional foods , Nutraceuticals and Speciality foods with examples.	4
4.	1	Quality: Basic concepts. Various definitions of Quality, Quality Attributes of foods. Quailty Control Cycle. Causes of undesirable changes leading to quality deterioration in foods and their implications. Names of Regulatory and Accreditation Bodies.	4
	2	Basic idea of Sensory attributes of Foods and their assessment.	4
5.	1	Basic concept of Food Security and Food safety, Adulteration in Foods, Concept of HACCP. Definition of Safe and Unsafe foods as per FSSAI. Toxicity in Foods. Basic idea of Names and Significance of of ISO series.	4
	2	Basic concepts of Food Packaging. Desirable attributes of Food Package, Importance of Label and its declaration. An understanding of commonly used terms as Shelf Life, Net weight, Gross Weight, Drained Weight, Expiry Date, Best Before etc.	4 42

SUBJECT CODE: BFT303	COURSE TITLE: BASICS FOOD MICROBIOLOGY
	SEMESTER: III
	PRE-REQUISITE : NIL

OBJECTIVE: To provide the students an understanding of microorganisms and their role in food spoilage, foodborne illness, along with a knowledge of beneficial role of microorganisms and an idea of Food Plant Hygiene control.

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Sources of microbial contamination in foods and understand the growth requirements of common food borne pathogens and spoilage microorganisms.
- Understand intrinsic aspects of foods and how they can be modified to control the growth of microorganisms in foods.
- Microbial spoilage in different types of foods and its prevention and control
- To identify and control food borne diseases and food spoilage

S.No	Name of Authors / Books / Publishers	Year of Publication / Reprint
1	Frazier J & Westhoff DC, " Food Microbiology", 18th Edition, Tata McGraw Hill	2003
2	Garbutt J," Essentials of Food Microbiology" , 2nd Edition, Hodder Arnold , UK	1997
3	Pelczar, Chan and Krieg, Microbiology, 4th Edition, TMHhttps://www.academia.edu/60017426/Microbiology_by_pelczar_chan_and_krieg	1977
4	Ray B., " Fundamentals of Food Microbiology", 5th Edition, CRC Press	2005
5	Prescott, L.M., Harley, J.P. & Klein, D.A. Microbiology. 10th Edition. Tata McGraw-Hill	2017

COURSE DETAILS:

Units	S. No.	Contents	Lecture Hours
1.	1	General characteristics of microorganism. Classification, morphology, physiology, growth, nutrition and reproduction of microorganisms. Spores.	4
	2	Microbial Growth Curve & its implications. Pure culture techniques and maintenance of cultures	4
2.	1	Incidence of microorganism in foods, Sources of contamination. Principles underlying spoilage of foods. Control of microorganisms.	4
	2	Physical and Chemical methods to control microorganisms. Food poisoning and food borne infections, Microbial toxins, Indicator organisms.	4
3.	1	Sources of Contamination and spoilage of cereal products & sugar products . Methods adopted for control and preservation	4
	2.	Sources of Contamination and spoilage of fruit and vegetable products, Bakery Products, Methods adopted for control and preservation	4
4.	1	Sources of Contamination and spoilage of Meat products, Fish and Sea foods, Egg and Poultry products. Methods adopted for control and preservation	4
	2	Sources of Contamination and spoilage of Milk and Milk products. Methods adopted for control and preservation.	4
5.	1	Microbes in food fermentation, putrefaction, Lipolysis. Antagonism and Synergism in microorganisms	4
	2	Food plant sanitation, inspection and control, personnel hygiene. Microbiological limits.	4

SUBJECTCODE: BFT351	**COURSE TITLE: ENERGY & ENVIRONMENT LAB**
	SEMESTER: III
	PRE REQUISITES: NIL

OBJECTIVE:
To make students understand the basic fundamental principles of energy engineering by performing different experiments and to make them correlate theory and the practical process by experimentation.

COURSEOUTCOME: On completion of this course ,the students will be able to

- Analyze the properties of liquid fuels.
- Analyze the various parameters for water and wastewater samples.
- Study solar energy apparatus.

S.No.	NAME OF EXPERIMENTS
1.	To determine flash and fire point – Pensky Martin apparatus
2.	To determine the viscosity of oil – Saybolt/Redwood viscometer
3.	To determine the Calorific Value of a given Liquid Fuel
4.	To determine the Moisture Content in a given coal sample.
5.	To determine the Volatile Content in a given coal sample.
6.	To determine the pH of a given water sample
7.	To determine the alkalinity of a given water sample
8.	To determine the total dissolved solids of the given water sample.
9.	Study of Solar Thermal Heater
10.	Study of Solar lamp.

SUBJECT CODE: BFT352	COURSE TITLE: ANALYSIS OF FOOD CONSTITUENTS LAB
	SEMESTER: III
	PRE-REQUISITE : NIL

OBJECTIVE: To study the application and analytical procedures for characterizing the properties of foods and their constituents. These analytical procedures are used to provide information about a wide variety of different characteristics of foods, including their composition, structure, physicochemical properties and sensory attributes.

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Gain knowledge of practices for evaluation of appropriate methods for analysis of foods
- Understand various methodologies for analysis of components in foods.
- Knowledge of procedures for sampling, sample storage, preparation
- Knowledge of Good Laboratory Practices and proper record keeping.
- Learn about common instrumentation used in food analysis.

S.No	Name of Authors / Books / Publishers	Year of Publication / Reprint
1	Pomeranz Y., Food Analysis Theory and Practice 3ed, CBS Publishers & Distributors Pvt. Ltd.	2004
2	S. Suzanne Nielsen , Introduction to Food Analysis, Springer 4th Edition https://fcen.uncuyo.edu.ar/upload/food-analysis.pdf	2010
3	R. E. Wrolstad, T. E. Acree, Eric A. Decker, M. H. Penner, D. S. Reid, S J. Schwartz, C. F. Shoemaker, D. Smith, Peter Sporns, Handbook of Food Analytical Chemistry, Wiley Online Library. https://onlinelibrary.wiley.com/doi/book/10.1002/0471709085	2004
4	Manuals of Methods of Analysis of foods. FSSAI, New Delhi https://fssai.gov.in/upload/uploadfiles/files/Manual_Oil_Fat_25_05_2016(1).pdf	2016

COURSE DETAILS:

Contents
1. Sampling requirements, procedures and methods .
2. Determination of moisture content of foods by oven drying method.
3. Determination of Total and Acid insoluble ash content in foods.
4. Determination of Crude fat content by solvent extraction method in foods.
5. Determination of crude Protein by Kjeldhal / Lowry methods.
6. Determination of reducing and total sugar content in foods.
7. Determination of crude Fibre content in foods
8. Determination of specific mineral contents in foods such as Ca, Iron, P, Chloride etc.
9. Determination of specific vitamin content of food such as ascorbic acid.
10. Chromatographic Separation and identification of amino acids.

SUBJECT CODE: BFT353	COURSE TITLE: BASIC FOOD MICROBIOLOGY LAB
	SEMESTER: III
	PRE-REQUISITE : NIL

OBJECTIVE: This is aimed to provide a hands-on opportunity for the students to practice and observe the principles of food microbiology. Students will familiarize themselves with the techniques used to research, regulate, prevent, and control the microorganisms found in food and understand the function of beneficial microorganisms during the food manufacturing process

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Obtain a good understanding of the microscope, microbial stains and culture media used in microbiology.
- Obtain knowledge on how different processing, storage and environmental conditions affect microbial changes in food systems.
- Obtain knowledge on and how to adjust these conditions to improve or minimize microbial deterioration of foods systems

S.No	Name of Authors / Books / Publishers	Year of Publication / Reprint
1	Manual of Methods of Analysis of Foods- Microbiological Testing. Lab Manual 14. FSSAI, GoI, New Delhi.	2012
2	Manual of Methods of Analysis. Microbiological Examination of Food and Water. https://www.fssai.gov.in/upload/uploadfiles/files/Manual-Microbiology-Methods.pdf	2015
3	Osman Erkmen, Basic Methods for the Microbiological Analysis of Foods. 3rd Edition, Nobel Publishing.	2015
4	Yousef,A.L.,Food Microbiology :A Laboratory Manual.Wiley Inter-Science New Jersey.	2003

COURSE DETAILS:

Contents
1. Microscope its parts and utility in identification and differentiation of various microorganisms as bacteria, yeast and mold.
2. Micrometry and determination of size of different microbes.
3. Preparation and sterilization of media and glass ware for microbial counts.
4. Familiarization with common techniques for handling pure culture, Serial dilution, Inoculation, slide preparation , Incubation etc.
5. Simple and differential staining of microorganisms and their examination.
6. Direct total, viable, and non-viable count of microorganisms in given food sample.
7. Determination of Standard Plate Count (SPC) in natural and/or processed foods.
8. Microbiological examination of some selected natural and processed foods.
9. Microbiological examination of potable water.
10. Enumeration of Coliforms organisms in some selected processed food samples

IV SEMESTER

SUBJECT CODE: BFT401	**COURSE TITLE: FLUID FLOW AND SOLID HANDLING**
	SEMESTER:III (ODD)
	PRE REQUISITES: NIL

OBJECTIVE:

- To present the fundamental insights of fluids and their static and dynamic behaviors and fluid machineries, etc.
- To impart knowledge on particle size analysis, size reduction, separation of solid particles from fluids and flow through porous media.

COURSEOUTCOME:

On completion of this course, the students will be able to

- Understand the properties and flow of fluid.
- Explain the factors influencing velocity profiles for laminar and turbulent flow.
- Design of pumps and compressors for optimum operation.
- Estimate the crushing efficiency of different type of crushers.
- Measure the particle size and understand the particle separation operation.

REFERENCE BOOKS

S.No.	NAMEOFAUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION /REPRINT
1.	McCabe Smith: Unit Operations in Chemical Engineering 7TH Edition, McGraw Hill	2017
2.	Fox, R.A. & McDonald, A.T. "Introduction to Fluid Mechanics, 5th ed: John Wiley	1998
3.	Kumar D S "Fluid Mechanics", S.K, Katria and Sons, Delhi	1998
4.	RajputR.K." Textbook of Fluid Mechanics", S.Chand and Co., New Delhi	1998
5.	Gupta, Vijay and SK Gupta, Fluid Mechanics and its Applications. New Age International Publishers	2015

Units	Contents	Lecture Hours
I.	**Solids and Their Handling:** Properties of solids,screening, industrial screening equipment. Determination of particle size, screen analysis,Sizereduction of solids, methods and laws of size reduction, operating variables, intermediate and fine size reduction, and Energyrelationshipsinsize reduction. Power-driven machines: Crushers, grinders, and conveyors.	10
II.	**Particle Separation:** Theory, continuous and batch equipment. Flow of solids through fluids, classification, Gravity settling, Sedimentation ;Thickening; Elutriation, Centrifugal separation, floatation, filtration and magnetic separation	8
III.	**Fluid flow:** Properties of fluids. Fluid statics: Euler's equation, Hydrostatic Law and Pressure Measurement. Transport of fluids, energy relationships, pipe fittings, head losses in pipe flow **Flow measurements**: Flow rate and velocity measurements, Orifice meter. Nozzle and venturi meter, rotameter and pitot tube.	12
IV.	**Pumping and compressing**: reciprocating pumps, rotary pumps, centrifugal pumps and blowers. Introduction to fluidization.	7
V.	**Agitation and Mixing:** Agitated vessels; Blending and mixing; Suspension of solid particles ;Dispersion operations; Agitator selection.	5

SUBJECT CODE: BFT402	COURSE TITLE: FOOD CHEMISTRY
	SEMESTER: III
	PRE-REQUISITE : NIL

OBJECTIVE: The course aims to provide systematic knowledge and understanding of chemistry of food components like Water, Proteins, Carbohydrates ,Lipids, Vitamins and Minerals. It covers understanding of the various reactions and interactions involved and that are significant to Food Quality.It also covers the aspect of various Food Additives and their role in natural/ processed foods..

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the chemistry of food components like water, proteins, carbohydrates and lipids.
- Understand the chemistry of Vitamins, Minerals and Pigments found in foods.
- Learn about the functional properties of food constituents that are of importance in processing& quality
- Learn about the food additives that are relevant to processed food industry for shelf life extension and as processing aids

S.No	Name of Authors / Books / Publishers	Year of Publication / Reprint
1	Shakuntala Manay,N. and Shadaksharaswamy, M. Food Facts and Principles. New Age International Publishers.	2013
2	Meyer, L.J., Food Chemistry. CBS Publishers & Distributors	2004
3	Damodaran S., Parkin K.L., editors of Fennema's Food Chemistry, 5th Edition, CRC Press, Taylor & Francis Group https://www.google.co.in/books/edition/Fennema_s_Food_Chemistry	2017
4	John M. deMan, Principles of Food Chemistry, 3rd Edition. Springer	2013

COURSE DETAILS:

Units	S. No.	Contents	Lecture Hours
1.	1	Water in Foods: Structure. Properties, Interactions, Water activity, Sorption Isotherms and food stability.	4
	2	Carbohydrates: Structure & Classification. Functions, Reactions and properties of simple and complex carbohydrates, Browning reactions, Modified Starches and their applications in food industry.	5
2.	1	Lipids: Classification and Structure. Consistency of commercial fats, Lipolysis, Auto oxidation, Thermal Decomposition of lipids. Antioxidants and their role.	4
	2	Refining of oils, Modification of oils and fats, Role of food lipids in flavor, Nutritional aspects of natural and modified fats. Medium Chain Triglycerides as Functional Foods.	4
3.	1	Proteins: Structure, Chemical reactions and interactions of amino acids and proteins. De-naturation and its implications, Functional properties of food proteins	5
	2.	Modification of food proteins in processing and storage and its implications. Nutritive value of food proteins.	4
4.	1	Vitamins and Minerals: Chemistry and stability of water and fat soluble vitamins, Chemical properties of minerals and their bioavailability, Enrichment Fortification and Restoration.	5
	2	Natural pigments in foods and their retention in processed foods. Flavoring constituents in foods, Development of process and reaction flavour volatiles.	4
5.	1	Food Additives: Definitions, sources, uses and functions and regulatory aspects of Food additives. Limits of usage as per FSSAI.	4
	2	Intentional & Non Intentional additives, Food Adulterants.	3

SUBJECT CODE: BFT403	COURSE TITLE: FOOD BIOCHEMISTRY & BIOTECHNOLOGY
	SEMESTER: IV
	PRE-REQUISITE : NIL

OBJECTIVE: To furnish a student with knowledge and understanding of the basic biological and chemical processes of living cells and enzymes and how these are harnessed into industrial processes and technologies for the production, processing and preservation of food and related products.

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the fundamental biochemical principles, structure/function of bio-molecules, metabolic pathways and the regulation of biochemical processes.
- Understand the mechanism of enzyme action and identify factors affecting their action.
- Understand and be able to control the major chemical and biochemical (enzymatic) reactions that influence food quality with emphasis on food industry applications.
- Understand how the properties of different food components and interactions among these components modulate the specific quality attributes of food systems.

S.No	Name of Authors / Books / Publishers	Year of Publication / Reprint
1	A L Lehninger, Principle of Biochemistry, 8th Edition, W H Freeman & Co.Ltd.	2021
2	Berg, J.M, Tymoczko J.M, Lubert Stryer," Biochemistry", 6th Edition, W H Freeman & Co.	2006
3	Shubhangini A. Joshi, Nutrition and Dietetics, 4thEdition. Tata McGraw Hill Publishing Company Ltd.	2017
4	JH Weil, " General Biochemistry", 6th Edition, New Age International Private Limited	1990

COURSE DETAILS:

Units	S. No.	Contents	Lecture Hours
1.	1	Enzymes: Nomenclature, classification, sources and properties, mechanism of action, coenzyme and prosthetic groups.	3
	2	Regulation of enzymatic activity, enzyme kinetics, Michaelis Menten equation, effects of pH & temperature, Allosteric enzymes, Inhibition & specificity of enzymes.	4
2.	1	Application of enzymes in food processing: Functions, importance and role in food industry. Relevant examples.	3
	2	Endogenous enzymes and their, sources, conversions and specific applications. Role of endogenous enzymes in modification of foods	3
3.	1	Mechanism of Digestion, Absorption, Assimilation and Transport of nutrients in human beings	3
	2	Metabolism of carbohydrates, lipids & proteins: glycolysis and respiration, production of ATP, brief description of electron transport chain, β oxidation of long chain fatty acids, ketosis, biosynthesis of fatty acids, Breakdown of proteins, transamination, deamination, decarboxylation, nitrogen fixation, Urea Cycle.	5
4.	1	Physiological development; Growth, Maturation and senescence, Fruit ripening and associated biochemical changes, Effect of these developments on quality of fresh produce.	3
	2.	Physiology & biochemistry of Respiration in fresh produce. Climacteric and non Climacteric fruits. Biochemical changes as a result of aerobic and anaerobic metabolism, during maturation / storage.	4
5.	1	Structure and development of muscle. Biochemical changes associated during conversion of muscle to meat. Post Mortem Glycolysis and Rigor-Mortis.	4
	2	Ante-mortem and post-mortem factors affecting quality of meat Biochemical reactions leading to Quality changes in composition, color, flavour and texture of meat.	3

SUBJECT CODE: BFT451	**COURSE TITLE: FLUID FLOW & SOLID HANDLING LAB**
	SEMESTER: IV
	PRE REQUISITES:NIL

OBJECTIVE: To provide practice in estimating friction losses , impart training to use various flow measuring devices and to determine the various parameters related to fluid flow in pipes and in open channels.

COURSE OUTCOME : On completion of this course, the students will be able to

- Estimate the friction and measure the frictional losses in fluid flow.
- Understand the functioning of various flow measurement devices as venturi meter& orifice meter
- Predict the coefficient of discharge for flow through pipes.

S.No.	NAME OF EXPERIMENTS
1	Study and verification of the flow pattern in Bernoulli's apparatus.
2	To determine the coefficient of discharge of the venturi meter.
3	To determine the coefficient of discharge of the orifice meter.
4	To study the minimum fluidization velocity and pressure drop for a liquid solid system.
5	Study of Pipe fittings and Valves.
6	To determine the average particle size of a mixture of particles by sieve analysis
7	To determine the size reduction ratio and critical speed by conducting an experiment in a ball mill.
8	To Study the settling characteristics of a given suspension
9	To study the filtration characteristics of the Rotary Vacuum /Plate and Frame filter.
10	To determine the efficiency of a cyclone separator.

SUBJECT CODE: BFT452	**COURSE TITLE: FOOD CHEMISTRY LAB**
	SEMESTER: IV
	PRE-REQUISITE : NIL

OBJECTIVE: To study the analytical procedures for characterizing the properties of foods constituents and their interactions that affect the quality and stability of foods.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Use the laboratory techniques common to basic and applied food chemistry.
- Understand the principles behind analytical technique when presented with a practical problem.
- Demonstrate practical proficiency in a food analysis laboratory.
- To learn about the various food reactions and interactions through experiments.

S.No	Name of Authors / Books / Publishers	Year of Publication / Reprint
1	Dennis D. Miller, C. K. Yeung. Food Chemistry: A Laboratory Manual, 2nd Edition. Wiley	2022
2	S. Suzanne Nielsen. Food Analysis Laboratory Manual. 3rd Edition. Springer International Publishing. https://edisciplinas.usp.br/pluginfile.php/5516437/course/section/6014434/2017_Book_FoodAnalysisLaboratoryManual.pdf	2019
3	Connie M. Weaver and James R. Daniel. The Food Chemistry Laboratory: A Manual for Experimental Foods, Dietetics and Food Scientists. 2nd Edition CRC PRESS. https://dl.icdst.org/pdfs/files/42dc13712c69228c56615b2fdfe70632.pdf	2005

COURSE DETAILS:

Contents
1. Analysis of water for potable and food purposes
2. Moisture content in foods in relation to their stability
3. Non-enzymatic browning reactions and its determination
4. Determination of rate/ extent of hydrolysis of sucrose/starch
5. Determination of free fatty acid content in fats and oils
6. Detection and estimation of oxidative rancidity in fats/oils (Peroxide, Acid, Anisidene Values)
7. Determination of heat stability of vitamin C
8. Study of some reactions of proteins
9. Study of some processing changes in proteins
10. Study of some functional properties of proteins

SUBJECT CODE: BFT453	COURSE TITLE: FOOD BIOCHEMISTRY LAB
	SEMESTER: IV
	PRE-REQUISITE : NIL

OBJECTIVE: To give the students a solid foundation in biology and chemistry and to develop analytical and critical thinking skills in biological phenomena through scientific methods.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Learn fundamental approaches for experimentally investigating biochemical problems
- Learn the enzyme kinetics and the effect of various factors on enzyme activity
- Understand the applicability of the biochemical methods to realistic situations

S.No	Name of Authors / Books / Publishers	Yearof Publication / Reprint
1	Wilson W. Principles and Techniques of Practical Biochemistry. Cambridge University Press, England.	2005
2	Dr. G. Sattanathan, Dr. S.S. Padmapriya, Ph.D. and Dr. B. Balamuralikrishnan. PracticalManual ofBiochemistry. Skyfox Publishing Group. https://skyfox.co/wp-content/uploads/2020/12/Practical-Manual-ofBiochemistry.pdf	2020
3	Nutritional Biochemistry Practical Manual. Indira Gandhi National Open University School of Continuing Education. https://egyankosh.ac.in/handle/123456789/93464	2020

COURSE DETAILS:

Contents
1. Determination of enzyme activity and specific activity (Enzyme assay).
2. Determination of effect of temperature on enzyme activity.
3. Determination of effect of pH on enzyme activity.
4. Determination of effect of substrate concentration on enzyme activity & estimation of Km value.
5. Estimation of enzymatic browning in foods.
6. Estimation of enhancement in an enzyme activity during ripening of fruits.
7. Estimation of enhancement in an enzyme activity during sprouting of grains.
8. Detection/ estimation of catalase and peroxidase activity in vegetable.